THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE

SCHOOL INFRASTRUCTURE, MINERALS AND MANUFACTURING ENGINEERING

DEPARTMENT OF MECHANICAL ENGINEERING

B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION

COURSE CODE: MEE 101 | **COURSE TITLE:** Engineering Drawing I

UNITS: 3 | **TIME ALLOWED:** 3 hours

Instruction: Answer **QUESTION 1** and **ANY OTHER THREE** questions

1a.) Draw the Isometric projection of the object shown in Fig. Q1. **[20 marks]**

1b.) Draw the Front view (in the direction of the arrow F), the side view, and the plan view of the object shown in FIG. Q1, using the first angle projection. **[20 marks]**

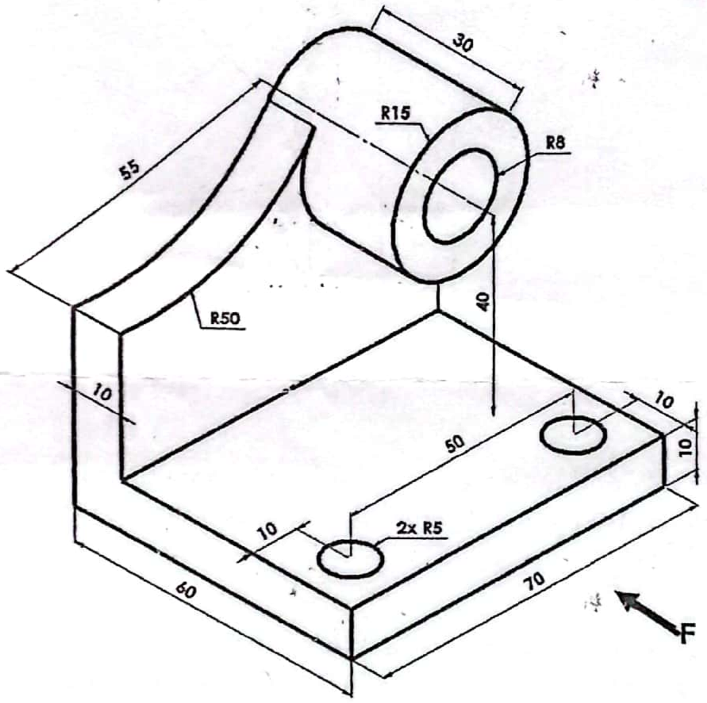

FIG Q1

2a.) A triangle PQR stands on the side PQ as base and has the following dimension: PQ is 89 mm, PR is 76 mm, angle RPQ is $52\frac{1}{2}°$. Circumscribe a circle on the triangle. Also inscribe a circle in the same triangle PQR. **[10 marks]**

2b.) Using the circumscribing rectangle method, construct a parabola with its axis vertical. The parabola has a rise of 128 mm and a span of 114 mm. **[10 marks]**

3a.) Construct regular quadrilateral, pentagon, hexagon, and octagon (together) having a common base of 30 mm. **[10 marks]**

3b.) Using the square method, construct a regular octagon with a distance across flats being 100 mm. **[10 marks]**

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4.) Reproduce FIG. Q4. All construction lines, dimensions and centers must be shown.

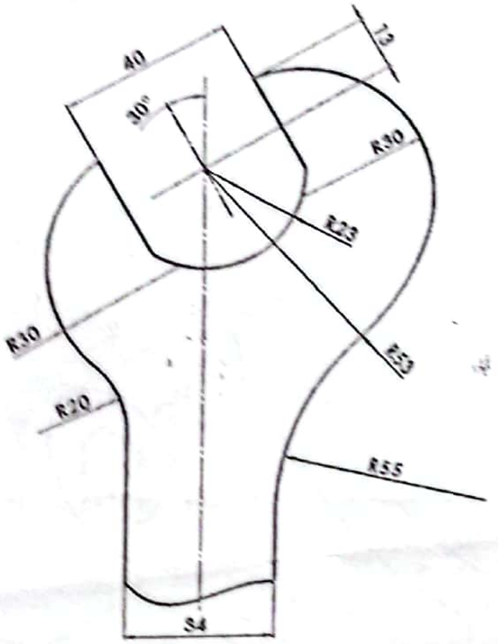

FIG Q4

[20 marks]

5.) An ellipse, parabola and hyperbola have a common focus 80 mm from the directrix. Draw all the three curves (together), given that the eccentricity of the ellipse is 2/5 and that of the hyperbola is 5/2. **[20 marks]**

6.) The crank and connecting rod of a steam engine are 30 mm and 100 mm long, respectively. The crank makes 270 rev/min in the clockwise direction. Fig. Q6 shows the instance of the mechanism when it has turned 60° from the inner dead centre. mechanism of a steam engine. Construct the locus of the point P for one revolution of the crank. end X describes an arc or a circle while the end Y describes a straight line.

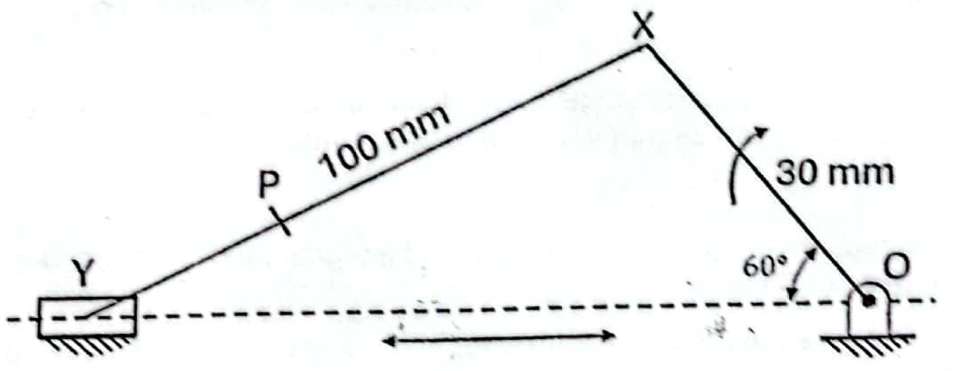

FIG Q6

[20 marks]